FinOps Buddy — AWS reference deployment architecture

Audience: Platform / security / FinOps engineers
Source: OpenSpec change `deploy-aws` (`openspec/changes/deploy-aws/proposal.md` , `design.md`)
Status: Proposed (not yet implemented in application code)

This document describes the **standard AWS architecture** agreed for running **FinOps Buddy** when internal users reach the app over **public HTTPS** while the data plane remains **sensitive** (multi-account cost and FinOps APIs). It complements local and Docker workflows (`~/.aws` profiles); production uses **IAM roles** and **OIDC** instead.

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1. Executive summary

Area	Choice
Compute	**Amazon ECS on Fargate** (long-lived process, agent/MCP warm-up, **SSE** streaming)
Edge	**AWS WAF** → **Application Load Balancer** (**HTTPS**, **OIDC** to corporate IdP)
Users	**Employees in browser**; **subset** via IdP app assignment + optional **group allowlist** in app
AWS data	**ECS task role** as hub; **STS AssumeRole** into **per-account FinOps reader** roles
LLM	**Amazon Bedrock** with **IAM** (preferred in production; model IDs are Bedrock-native, not OpenAI GPT names)
Compliance (v1)	**WAF logging**, **ALB access logs**, **application audit logs** (stable user id, no secrets)

Cognito is intentionally **out of scope** for v1 when the IdP supports **OIDC** directly.

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2. System context (diagram)

2.1 Edge, VPC runtime, Bedrock, and logging

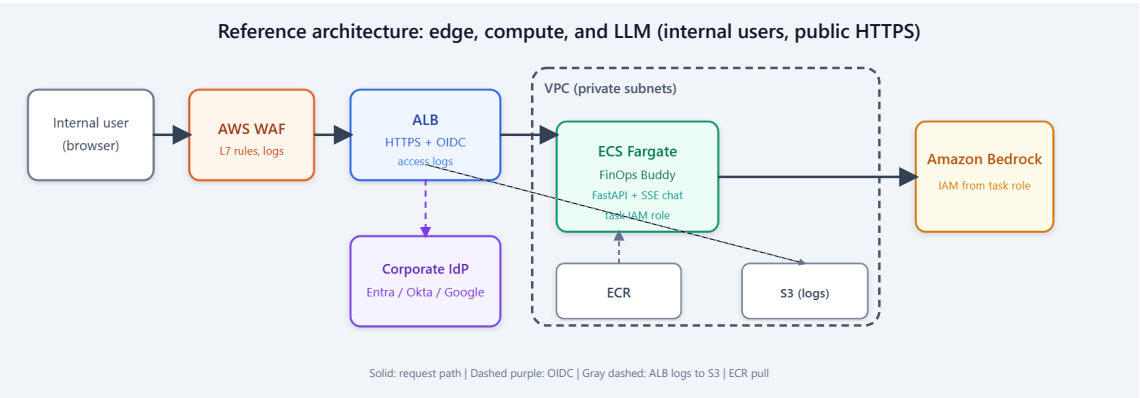

Figure: Traffic flow from browser through WAF and ALB (with OIDC to the corporate IdP) to ECS tasks; Bedrock invocation via task role; ECR pull and log delivery patterns.

2.2 Mermaid (for Markdown viewers that render Mermaid)

```
flowchart LR
  U[Internal user browser] --> W[AWS WAF]
  W --> A[ALB HTTPS + OIDC]
  A --> E[ECS Fargate FinOps Buddy]
  A -. OIDC .-> I[Corporate IdP]
  E --> B[Amazon Bedrock]
  E -. pull .-> R[ECR]
  A -. access logs .-> S[S3]
  W -. logs .-> S
```

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3. Identity and trust (diagram)

3.1 Three layers: people, application, AWS accounts

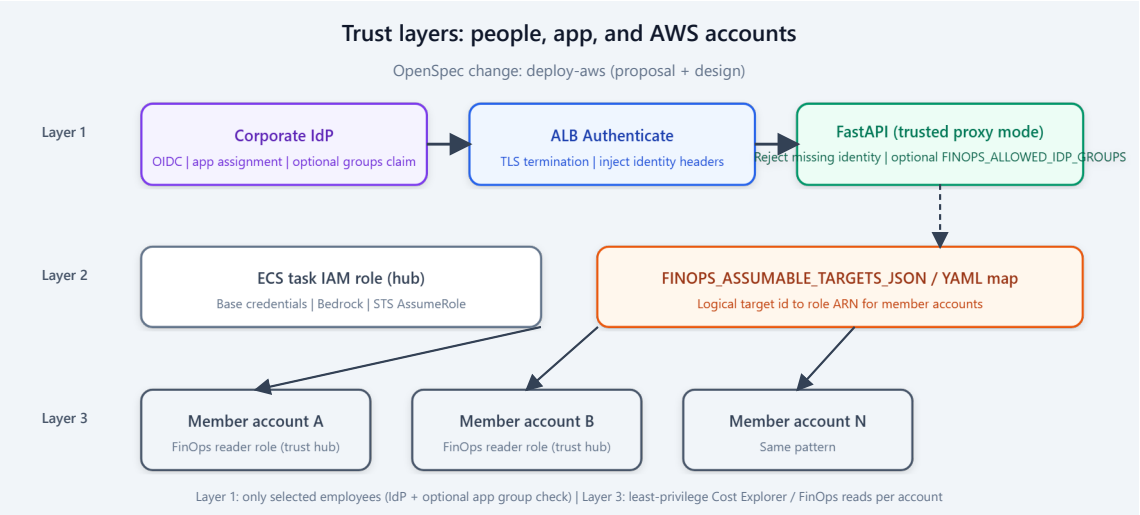

Figure: Layer 1 — IdP + ALB + optional FastAPI checks (`FINOPS_TRUSTED_PROXY_AUTH`, `FINOPS_ALLOWED_IDP_GROUPS`). Layer 2 — task role + target map (`FINOPS_ASSUMABLE_TARGETS_JSON` / YAML). Layer 3 — member-account reader roles trusting the hub role.

3.2 Mermaid

```
flowchart TB
  subgraph L1["Layer 1 - People"]
    IdP[Corporate IdP OIDC]
    ALB[ALB authenticate]
    APP[FastAPI trusted proxy mode]
  end
  IdP --- ALB
  ALB --> APP
  subgraph L2["Layer 2 - AWS hub"]
    TASK[ECS task IAM role]
```

```
    MAP[Target id to role ARN map]
  end
  APP --> TASK
  MAP --> TASK
  subgraph L3["Layer 3 - Member accounts"]
    A[Reader role account A]
    B[Reader role account B]
    N[Reader role account N]
  end
  TASK -->|AssumeRole| A
  TASK -->|AssumeRole| B
  TASK -->|AssumeRole| N
```

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4. Key design decisions (from `design.md`)

Decision	Why
Fargate	Fits streaming chat, warm-up, and WebSockets better than Lambda without a large refactor.
ALB OIDC	Standard integration with Entra / Okta / Google; fewer moving parts than Cognito for OIDC-first IdPs.
WAF on ALB	Layer-7 protection and logging for an internet-exposed admin-style application.
Sticky sessions (or single task initially)	**SSE** and long-lived connections need affinity when multiple tasks run.
AssumeRole per account	Aligns with least privilege and typical org SCP patterns; avoids one “god reader” unless policy allows.
Feature flags (`FINOPS_*` + YAML)	Local and Docker behavior stays default-off for cloud-only features.

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5. Configuration surface (planned)

Documented in OpenSpec `app-settings` delta; implementation will update **README** and **`config/settings.yaml`** .

Variable / area	Purpose
`FINOPS_CLOUD_DEPLOYMENT_MODE`	Enable cloud credential chain + target map instead of profile files.
`FINOPS_ASSUMABLE_TARGETS_JSON`	JSON list mapping logical target ids to **role ARNs** (replaces YAML map when set).
`FINOPS_TRUSTED_PROXY_AUTH`	Require ALB-injected identity headers; reject otherwise.
`FINOPS_ALLOWED_IDP_GROUPS`	Optional comma-separated allowlist intersected with IdP **groups** claim.
`FINOPS_LLM_PROVIDER`	e.g. `bedrock` to force Bedrock even if an OpenAI key exists.

Exact header names for OIDC claims SHALL be taken from **current AWS documentation** for ALB authenticate and mapped in one module during implementation.

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6. Observability and compliance (v1)

Signal	Use
ALB access logs	HTTP-level audit trail (URI, status, latency, source IP).
WAF logs	Block/allow decisions and rule matches.
IdP sign-in logs	Account lifecycle and MFA (outside this repo).
Application logs	Stable user identifier per allowed request; **never** log raw tokens or `Authorization` headers.

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7. Rollout and rollback (summary)

1. Ship application changes with **defaults off** (no change for existing users).
2. Deploy **staging**: IaC, ACM, OIDC app registration, cross-account trust.
3. Enable **trusted proxy auth** and optional group allowlist; validate **dashboard** and **SSE chat**.
4. **Production**: tighten IdP assignment, confirm log retention.

Rollback: Turn off `FINOPS_TRUSTED_PROXY_AUTH`, scale service to zero, or break-glass ALB rule deny-all.

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8. Diagram sources and PDF export

- **AWS Diagram MCP** (`generate_diagram`, Python `diagrams` package) was attempted for official AWS icon PNGs; on **Windows** the server hit path/parser errors (`\U` escapes in workspace paths). This document therefore uses **hand-authored SVG** files under `docs/diagrams/` (AWS-style layout, not trademarked icon assets).

- **PDF:** from the repo root run:

  ```
  poetry run python docs/build_deploy_architecture_pdf.py
  ```

  This uses **xhtml2pdf** (already in the project) so PDF generation works without WeasyPrint/GTK. On Linux/macOS with GTK installed, you may use **WeasyPrint** instead:

  ```
  FINOPS_USE_WEASYPRINT=1 poetry run python docs/build_deploy_architecture_pdf.py
  ```

  Output: `docs/DEPLOY_AWS_ARCHITECTURE.pdf`. Mermaid blocks in the Markdown are omitted from the PDF (SVG figures carry the same intent).

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9. Related artifacts

Artifact	Path
Proposal	`openspec/changes/deploy-aws/proposal.md`

Design	`openspec/changes/deploy-aws/design.md`
Specs	`openspec/changes/deploy-aws/specs/`
Tasks	`openspec/changes/deploy-aws/tasks.md`

Generated documentation aligned with the `deploy-aws` OpenSpec proposal. Implementation is tracked in `tasks.md`.